

BRAIN team

IMT Atlantique · January 2027



Lecture Plan

PART 1

Architectural Limits of Dense Attention

The Scaling Frontier: Sparsity & Alternative Attention

- **Dense Transformer constraints:** Every parameter is activated for every token ($\mathcal{O}(N)$ compute cost), and KV-cache footprint grows linearly with context length T .
- **Sparse Mixture-of-Experts (MoE):** Route each token to Top- k of E specialized MLP experts. Dramatically expands parameter capacity without increasing active FLOPs.
- **Recurrence & State Space Models:** Mamba / RWKV replace softmax attention with linear recurrent states, achieving $\mathcal{O}(1)$ inference memory per token.

